

Lecture Note 1: Probability Review

14.380: Statistical Methods in Economics, Fall 2026

In these notes, we review key probability concepts that will be used throughout the course. This material will only be briefly covered in lecture. If this material is unfamiliar to you, you are encouraged to read these notes carefully and supplement them with the suggested readings.

1 Principles of Probability

A random experiment is an experiment whose outcome cannot be predicted beforehand. How do we model a random experiment? There are three key elements: the sample space, the events and the probability measure. Each of these are described in turn.

The *sample space* Ω is the set of all possible outcomes of a random experiment. We denote an outcome as $\omega \in \Omega$. An *event* A is a subset of the sample space, $A \subseteq \Omega$.

Example 1.

1. Suppose we survey 10 randomly selected people on their employment status and count how many are unemployed: $\Omega = \{0, 1, 2, \dots, 10\}$. A is the event that more than 30% of those surveyed are unemployed: $A = \{4, 5, 6, \dots, 10\}$.
2. Suppose we ask a random person what is their income: $\Omega = \mathbb{R}_+$. A is the event that the person earns between \$30,000 and \$40,000: $A = [30,000, 40,000]$.

We place additional restrictions on collections of events $\mathcal{A} \subseteq 2^\Omega$. These impose sufficient structure that will allow us to consistently define the probabilities of events.

Definition 1. Let Ω be a set and $\mathcal{A} \subseteq 2^\Omega$ be a family of its subsets. $\mathcal{A}$ is a σ -algebra if and only if it satisfies the following:

- (i) $\Omega \in \mathcal{A}$;
- (ii) $\mathcal{A}$ is closed under complements: $A \in \mathcal{A}$ implies that $A^C = \Omega - A \in \mathcal{A}$;
- (iii) $\mathcal{A}$ is closed under countable union: If $A_n \in \mathcal{A}$ for $n = 1, 2, \dots$, then $\cup_n A_n \in \mathcal{A}$.

We say $(\Omega, \mathcal{A})$ is a *measurable space* and $A \in \mathcal{A}$ is *measurable* with respect to the σ -algebra $\mathcal{A}$.

Notice that Properties (i) and (ii) imply $\emptyset \in \mathcal{A}$. Properties (ii) and (iii) imply $\mathcal{A}$ is closed under countable intersection by DeMorgan's Law. That is, if $A_n \in \mathcal{A}$ for $n = 1, 2, \dots$, then $\cap_n A_n \in \mathcal{A}$.

Definition 2. Let $(\Omega, \mathcal{A})$ be a measurable space. A *measure* is a function, $\mu: \mathcal{A} \rightarrow \mathbb{R}$ such that

- (i) $\mu(\emptyset) = 0$;
- (ii) $\mu(A) \geq 0$ for all $A \in \mathcal{A}$;

(iii) If $A_n \in \mathcal{A}$ for $n = 1, 2, \dots$ with $A_i \cap A_j = \emptyset$ for $i \neq j$, then

$$\mu\left(\bigcup_n A_n\right) = \sum_n \mu(A_n).$$

If $\mu(\Omega) < \infty$, μ is a *finite measure*. If $\mu(\Omega) = 1$, μ is a *probability measure*. We denote a probability measure as $P: \mathcal{A} \rightarrow [0, 1]$.

Definition 3. A *probability space* is a triple $(\Omega, \mathcal{A}, P)$, where Ω is a set, $\mathcal{A}$ is a σ -algebra and P is a probability measure on $\mathcal{A}$.

We've now defined all components needed to model a random experiment. A random experiment is characterized by its probability space, $(\Omega, \mathcal{A}, P)$. With the definition of a probability space that we laid out above, we are ready to state the usual probability facts.

Theorem 1. Consider a probability space $(\Omega, \mathcal{A}, P)$. The following hold:

1. For all $A \in \mathcal{A}$, $P(A^C) = 1 - P(A)$.
2. If $A_1, A_2 \in \mathcal{A}$ with $A_1 \subseteq A_2$, then $P(A_1) \leq P(A_2)$.
3. For all $A \in \mathcal{A}$, $0 \leq P(A) \leq P(\Omega)$.
4. If $A_1, A_2 \in \mathcal{A}$, then $P(A_1 \cup A_2) = P(A_1) + P(A_2) - P(A_1 \cap A_2)$.

Probability measures are also *continuous* along monotone sequences of events.

Theorem 2 (Continuity of probability measures). Consider a probability space $(\Omega, \mathcal{A}, P)$ and let $A_n \in \mathcal{A}$ for $n = 1, 2, \dots$

1. If $A_1 \subseteq A_2 \subseteq \dots$, then $\lim_{n \rightarrow \infty} P(A_n) = P(\cup_n A_n)$.
2. If $A_1 \supseteq A_2 \supseteq \dots$, then $\lim_{n \rightarrow \infty} P(A_n) = P(\cap_n A_n)$. In particular, if $\cap_n A_n = \emptyset$, then $\lim_{n \rightarrow \infty} P(A_n) = 0$.

Proof. Consider the first statement. Define $B_1 = A_1$ and $B_n = A_n - A_{n-1}$ for $n \geq 2$. The events B_n are disjoint, lie in $\mathcal{A}$, and satisfy $\cup_{n=1}^N B_n = A_N$ for each N as well as $\cup_n B_n = \cup_n A_n$. By countable additivity,

$$P(\cup_n A_n) = \sum_n P(B_n) = \lim_{N \rightarrow \infty} \sum_{n=1}^N P(B_n) = \lim_{N \rightarrow \infty} P(A_N).$$

The second statement follows by applying the first statement to the increasing sequence A_n^C and using $P(A^C) = 1 - P(A)$. $\square$

1.1 Conditional Probability

Conditional probability gives us a way to model the outcome of a random experiment conditional on some partial information. For instance, given a random experiment and the information that event B has occurred, what is the probability that the outcome also belongs to event A ? To do so, we define a new probability measure on Ω .

Definition 4. Let $A, B \in \mathcal{A}$ with $P(B) > 0$. The **conditional probability of A given B** is

$$P(A | B) = \frac{P(A \cap B)}{P(B)}.$$

Observe that $P(A | B)$ is also a probability measure. We can therefore think of $P(A | B)$ as part of a new probability space with $\Omega = B$ and $P(S) = P(S | B)$ for any $S \subseteq B$. Furthermore, the definition of conditional probability implies the following useful formula. We have that

$$P(A \cap B) = P(A|B)P(B).$$

We next list several important results about conditional probabilities (stated again without proof).

Theorem 3.

1. Multiplication rule:

$$P(\cap_{i=1}^n A_i) = P(A_1)P(A_2 | A_1)P(A_3 | A_2 \cap A_1) \dots P(A_n | \cap_{i=1}^{n-1} A_i).$$

2. Law of total probability: Consider K disjoint events C_k that partition Ω . That is, $C_i \cap C_j = \emptyset$ for all $i \neq j$ and $\cup_{i=1}^K C_i = \Omega$. Let C be some event. Then,

$$P(C) = \sum_{i=1}^K P(C | C_i)P(C_i).$$

3. Bayes' Rule:

$$P(B | A) = \frac{P(A | B)P(B)}{P(A | B)P(B) + P(A | B^C)P(B^C)}.$$

Independence: As mentioned, we use conditioning to describe the partial information that an event B gives about another event A . What if B provides no information about A ?

Definition 5. Two events A, B are **independent** if $P(A | B) = P(A)$ and $P(B | A) = P(B)$. Equivalently, they are *independent* if $P(A \cap B) = P(A)P(B)$.

We can extend the definition of independence to collections of events.

Definition 6. Let $E_1, \dots, E_n$ be events. $E_1, \dots, E_n$ are **jointly independent** if for every subcollection of distinct indices $i_1, \dots, i_k$,

$$P(E_{i_1} \cap \dots \cap E_{i_k}) = P(E_{i_1}) \dots P(E_{i_k}).$$

Equivalently, whenever the conditioning event has positive probability,

$$P(E_{i_1} \mid E_{i_2} \cap \dots \cap E_{i_k}) = P(E_{i_1}).$$

Moreover, since conditional probabilities are probability measures, we can define independence with respect to a conditional probability.

Definition 7. Given an event C , events A, B are **conditionally independent** if

$$P(A \cap B \mid C) = P(A \mid C)P(B \mid C).$$

2 Random Variables

Consider a probability space $(\Omega, \mathcal{A}, P)$. For our purposes, a *random variable* will be defined to be a function from the outcome space Ω to the real-line $\mathbb{R}$: that is, $X: \Omega \rightarrow \mathbb{R}$.¹

Definition 8. Let X be a random variable. The *cumulative distribution function* (cdf) $F_X: \mathbb{R} \rightarrow [0, 1]$ of X is defined as

$$F_X(x) = P(X^{-1}((-\infty, x])) = P(\{\omega \in \Omega : X(\omega) \leq x\})$$

For simplicity, we often write

$$F_X(x) = P(X \leq x)$$

The cumulative distribution function F_X has the following properties:

1. For $x_1 \leq x_2$,

$$F_X(x_2) - F_X(x_1) = P(x_1 < X \leq x_2);$$

2. $\lim_{x \rightarrow -\infty} F_X(x) = 0$, $\lim_{x \rightarrow \infty} F_X(x) = 1$;

3. $F_X(x)$ is non-decreasing;

4. $F_X(x)$ is right-continuous: $\lim_{x \rightarrow x_0^+} F_X(x) = F_X(x_0)$.

Discrete random variables: A *discrete random variable* X is characterized by a list of possible values, $\mathcal{X} = \{x_1, \dots, x_n\}$, and their probabilities, $p = \{p_1, \dots, p_n\}$, where p_i denotes the probability that X will take value x_i , i.e. $p_i = P\{X = x_i\}$ for all $i = 1, \dots, n$. Note that $p_1 + \dots + p_n = 1$

¹More formally, we would need to be careful to define X such that the events that give rise to specific real values lie in $\mathcal{A}$ — i.e., we need X to be *measurable* with respect to $\mathcal{A}$. This level of precision is not needed for our purposes.

and $p_i \geq 0$ for all $i = 1, \dots, n$ by definition of probability. Then the cdf of X is given by $F_X(t) = \sum_{j=1, \dots, n: x_j \leq t} p_j$.

The *probability mass function* (pmf) of X is defined as

$$f_X(x) = \begin{cases} p_i & \text{if } x = x_i, \quad i = 1, 2, \dots \\ 0 & \text{otherwise} \end{cases}$$

Continuous random variables: A *continuous random variable* X is characterized by its *probability density function* (pdf), $f_X : \mathbb{R} \rightarrow \mathbb{R}$, such that $P\{a < X \leq b\} = \int_a^b f_X(s)ds$. Note that $\int_{-\infty}^{+\infty} f_X(s)ds = 1$ and $f_X(s) \geq 0$ for all $s \in \mathbb{R}$ by definition of probability. Then the cdf of X is given by $F_X(t) = \int_{-\infty}^t f_X(s)ds$. By the Fundamental Theorem of Calculus, $f_X(t) = dF_X(t)/dt$. We call $S_X = \{x : f_X(x) > 0\}$ the *support* of X .

Mixed random variables: A random variable is referred to as *mixed* if it is not discrete and not continuous.

Quantiles: If cdf F of some random variable X is strictly increasing and continuous then it has inverse, $q(x) = F^{-1}(x)$. It is defined for all $x \in (0, 1)$. Note that

$$P\{X \leq q(x)\} = P\{X \leq F^{-1}(x)\} = F(F^{-1}(x)) = x$$

for all $x \in (0, 1)$. Therefore $q(x)$ is called the *x-quantile* of X . It is such a number that random variable X takes a value smaller or equal to this number with probability x .

If F is not strictly increasing or continuous, then we define $q(x)$ as a generalized inverse of F , i.e. $q(x) = \inf\{t \in \mathbb{R} : F(t) \geq x\}$ for all $x \in (0, 1)$. In other words, $q(x)$ is a number such that $F(q(x) + \varepsilon) \geq x$ and $F(q(x) - \varepsilon) < x$ for any $\varepsilon > 0$.

Functions of random variables: Suppose we have random variable X and function $g : \mathbb{R} \rightarrow \mathbb{R}$. Then we can define another random variable $Y = g(X)$. The cdf of Y can be calculated as follows

$$F_Y(t) = P\{Y \leq t\} = P\{g(X) \leq t\} = P\{X \in g^{-1}(-\infty, t]\},$$

where g^{-1} may be the set-valued inverse of g . The set $g^{-1}(-\infty, t]$ consists of all $s \in \mathbb{R}$ such that $g(s) \in (-\infty, t]$, i.e. $g(s) \leq t$.

If g is strictly increasing and continuously differentiable then it has strictly increasing and continuously differentiable inverse g^{-1} defined on set $g(\mathbb{R})$. In this case $P\{X \in g^{-1}(-\infty, t]\} = P\{X \leq g^{-1}(t)\} = F_X(g^{-1}(t))$ for all $t \in g(\mathbb{R})$. If, in addition, X is a continuous random variable,

then

$$\begin{aligned}
f_Y(t) &= \frac{dF_Y(t)}{dt} = \frac{dF_X(g^{-1}(t))}{dt} \\
&= \left(\frac{dF_X(s)}{ds} \right) \Big|_{s=g^{-1}(t)} \left(\frac{dg(s)}{ds} \right)^{-1} \Big|_{s=g^{-1}(t)} \\
&= f_X(g^{-1}(t)) \left(\frac{dg(s)}{ds} \right)^{-1} \Big|_{s=g^{-1}(t)}
\end{aligned}$$

for all $t \in g(\mathbb{R})$. If $t \notin g(\mathbb{R})$, then $f_Y(t) = 0$.

One important type of function is a linear transformation. If $Y = X - a$ for some $a \in \mathbb{R}$, then

$$F_Y(t) = P\{Y \leq t\} = P\{X - a \leq t\} = P\{X \leq t + a\} = F_X(t + a).$$

In particular, if X is continuous, then Y is also continuous with $f_Y(t) = f_X(t + a)$. If $Y = bX$ with $b > 0$, then

$$F_Y(t) = P\{bX \leq t\} = P\{X \leq t/b\} = F_X(t/b).$$

In particular, if X is continuous, then Y is also continuous with $f_Y(t) = f_X(t/b)/b$.

Example 2. Suppose $X \sim U[0, 1]$ and $Y = X^2$. Over the support of X , this is a one-to-one transformation. We have that

$$X = \sqrt{Y}, \quad dX/dY = \frac{1}{2}y^{-1/2}.$$

Applying the formula above, we have that $S_Y = [0, 1]$ and

$$f_Y(y) = \frac{1}{2}y^{-1/2}.$$

2.1 Expectations

Informally, the *expectation* or *expected value* of some random variable can be interpreted as its average. Formally, if X is a random variable and $g : \mathbb{R} \rightarrow \mathbb{R}$ is some function, then, by definition,

$$\mathbb{E}[g(X)] = \sum_i g(x_i)p_i$$

for discrete random variables and

$$\mathbb{E}[g(X)] = \int_{-\infty}^{+\infty} g(x)f_X(x)dx$$

for continuous random variables.

Expected values for some functions g deserve special names:

- mean: $g(x) = x$, $\mathbb{E}[X]$

- second moment: $g(x) = x^2, \mathbb{E}[X^2]$
- variance: $g(x) = (x - \mathbb{E}[X])^2, \mathbb{E}[(X - \mathbb{E}[X])^2]$
- k -th moment: $g(x) = x^k, \mathbb{E}[X^k]$
- k -th central moment: $\mathbb{E}[(X - \mathbb{E}[X])^k]$

The *variance* of random variable X is commonly denoted by $V(X)$.

Expectations satisfy some important properties. We list them in the following result:

Theorem 4.

1. For any constant a (non-random), $\mathbb{E}[a] = a$.
2. Linearity: if X and Y are two random variables and a and b are two constants, then $\mathbb{E}[aX + bY] = a\mathbb{E}[X] + b\mathbb{E}[Y]$.
3. If X is a random variable, then $V(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$.
4. If X is a random variable and a is a constant, then $V(aX) = a^2V(X)$ and $V(X+a) = V(X)$.

Finally, we can use the expectation operator to express probabilities. An *indicator function* $I\{A\}$ is a function that is equal to one if condition A is true and zero otherwise. For example, if X is a random variable, then

$$I\{X \leq x\} = \begin{cases} 1 & \text{if } X \leq x \\ 0 & \text{otherwise} \end{cases}$$

Note that, for the continuous case,

$$\begin{aligned} \mathbb{E}[I\{X \leq x\}] &= \int_{-\infty}^{\infty} I\{X \leq x\} f_X(x) dx \\ &= \int_{-\infty}^x f_X(x) dx \\ &= F_X(x) = P(X \leq x). \end{aligned}$$

More generally, if $A_X \subseteq \mathbb{R}$, we have that

$$\mathbb{E}[I\{X \in A_X\}] = P(X \in A_X)$$

2.2 Examples of Random Variables

Discrete random variables:

- X is a discrete random variable that can only take on two values: 0, 1. We write $f_X(1) = p$, $f_X(0) = 1 - p$ and so,

$$f_X(x) = p^x(1 - p)^{1-x}.$$

Note that

$$\begin{aligned}\mathbb{E}[X^k] &= p, \quad k \geq 1, \\ V(X) &= p(1-p).\end{aligned}$$

We say that X has a *Bernoulli distribution*, and we write $X \sim \text{Bernoulli}(p)$.

- Suppose that X_i for $i = 1, \dots, n$ are i.i.d. Bernoulli random variables with $P(X_i = 1) = p$. Define $X = \sum_{i=1}^n X_i$. We say that X follows a *binomial distribution* with parameters n and p , and we write $X \sim \text{Binom}(n, p)$. X takes on values $0, 1, 2, \dots, n$. Its pmf is

$$f_X(x) = \binom{n}{x} p^x (1-p)^{n-x}$$

and

$$\mathbb{E}[X] = np, \quad V(X) = np(1-p).$$

- Suppose that X is a discrete random variable and takes on values $0, 1, 2, \dots$. Its pmf is

$$f_X(x) = \frac{\lambda^x e^{-\lambda}}{x!}, \quad \lambda > 0$$

We say that X is a *Poisson* random variable with parameter $\lambda > 0$ and we write $X \sim \text{Poisson}(\lambda)$. Note that

$$\mathbb{E}[X] = \lambda, \quad V(X) = \lambda.$$

Continuous random variables:

- Suppose that X is a continuous random variable with $f_X(x) = \frac{1}{b-a}$ for $x \in [a, b]$ and 0 otherwise. We say that X is *uniformly distributed* on $[a, b]$ and write $X \sim U[a, b]$.
- Suppose that X is a continuous random variable with support over $\mathbb{R}_+$. X is *exponentially distributed* with parameter $\lambda > 0$ if

$$f_X(x) = \lambda e^{-\lambda x}.$$

We write $X \sim \text{Exp}(\lambda)$ and have that

$$\mathbb{E}[X] = \frac{1}{\lambda}$$

$$V(X) = \frac{1}{\lambda^2}$$

- Suppose Z is continuously distributed with support over $\mathbb{R}$. We say that Z follows a *standard*

normal distribution if

$$f_Z(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}z^2}$$

We write $Z \sim N(0, 1)$ where $\mathbb{E}[Z] = 0, V(Z) = 1$. The cdf of standard normal random variable is typically written as $\Phi(z) = P(Z \leq z)$ and its pdf is written as $\phi(z)$.

We say that $X \sim N(\mu, \sigma^2)$ if $f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2\sigma^2}(x-\mu)^2}$. Note that $\mathbb{E}[X] = \mu, V(X) = \sigma^2$ and $X = \mu + \sigma Z$, where $Z \sim N(0, 1)$.

- Let $Z_i \sim N(0, 1)$ i.i.d. for $i = 1, \dots, n$. Let

$$X = \sum_{i=1}^n Z_i^2.$$

We say that X is a *chi-squared* random variable with n degrees of freedom and write $X \sim \chi_n^2$. It follows that $\mathbb{E}[X] = n, V(X) = 2n$.

- Let $Y_1 \sim \chi_k^2, Y_2 \sim \chi_l^2$ with $Y_1 \perp Y_2$. Define

$$Q = \frac{Y_1/k}{Y_2/l}.$$

We say that Q follows an *F-distribution* with k, l degrees of freedom. We write $Q \sim F_{k,l}$.

- Let $Z \sim N(0, 1)$ and $Y \sim \chi_n^2$ with $Z \perp Y$. Define

$$T = Z/\sqrt{Y/n}$$

We say that T is *student t-distributed* with n degrees of freedom and write $T \sim t_n$. We have that

$$\begin{aligned} \mathbb{E}[T] &= 0 \\ V(T) &= \begin{cases} \frac{n}{n-2}, & \text{if } n > 2, \\ \infty, & n = 1, 2 \end{cases} \end{aligned}$$

3 Multivariate Distributions

Let X, Y be two scalar random variables. A *random vector* (X, Y) is a mapping from Ω to $\mathbb{R}^2$. The *joint cumulative distribution function* of X, Y is

$$F_{X,Y}(x, y) = P(X \leq x, Y \leq y).$$

We say that (X, Y) is a *discrete random vector* if

$$F_{X,Y}(x, y) = \sum_{u \leq x} \sum_{v \leq y} f_{X,Y}(u, v),$$

where $f_{X,Y}(x,y) = P(X = x, Y = y)$ is the *joint probability mass function* of (X, Y) . We say that (X, Y) is a *continuous random vector* if

$$F_{X,Y}(x,y) = \int_{-\infty}^x \int_{-\infty}^y f_{X,Y}(u,v) dv du,$$

where $f_{X,Y}(x,y)$ is the *joint probability density function* of (X, Y) . As in the univariate case,

$$f_{X,Y}(x,y) = \frac{\partial^2 F_{X,Y}(x,y)}{\partial x \partial y}$$

at the points of continuity of $F_{X,Y}$.

From the joint cdf of (X, Y) , we can recover the *marginal cdfs*. We have that

$$\begin{aligned} F_X(x) &= P(X \leq x) \\ &= P(X \leq x, Y \leq \infty) \\ &= \lim_{y \rightarrow \infty} F_{X,Y}(x,y). \end{aligned}$$

We can also recover the *marginal pdfs* from the joint pdf using

$$f_X(x) = \sum_y f_{X,Y}(x,y) \quad \text{if discrete,}$$

and

$$f_X(x) = \int_{S_Y} f_{X,Y}(x,y) dy \quad \text{if continuous.}$$

Consider the discrete case. Let x be such that $f_X(x) > 0$. Then, the *conditional pmf* of Y given $X = x$ is

$$f_{Y|X}(y|x) = \frac{f_{X,Y}(x,y)}{f_X(x)}.$$

This satisfies the following two properties:

$$\begin{aligned} f_{Y|X}(y|x) &\geq 0 \\ \sum_y f_{Y|X}(y|x) &= 1. \end{aligned}$$

That is, $f_{Y|X}(y|x)$ is a well-defined pmf for a discrete random variable. The *conditional cdf* of Y given $X = x$ is then

$$F_{Y|X}(y|x) = P(Y \leq y | X = x) = \sum_{v \leq y} f_{Y|X}(v|x).$$

Next, consider the continuous case. It is analogous. For any x such that $f_X(x) > 0$, the

conditional pdf of Y given $X = x$ is

$$f_{Y|X}(y|x) = \frac{f_{X,Y}(x,y)}{f_X(x)}.$$

Provided that $f_X(x) > 0$, this is a well-defined pdf for a continuous random variable. The *conditional cdf* is

$$F_{Y|X}(y|x) = \int_{-\infty}^y f_{Y|X}(v|x) dv.$$

3.1 Conditional Expectations

Given a pair of random variables (X, Y) with a joint density $f_{X,Y}(x, y)$, we can define the *conditional expectation* of Y given $X = x$ as

$$\mathbb{E}[Y|X = x] = \int_{S_Y} y f_{Y|X}(y|x) dy.$$

Note that this is a function of x . It is sometimes denoted $\mu_Y(x)$ and called the *regression function*. In particular, this means we can view this as a random function $\mathbb{E}[Y|X]$. The following theorem is extremely useful.

Theorem 5 (Iterated expectations).

$$\mathbb{E}_Y[Y] = \mathbb{E}_X[\mathbb{E}_{Y|X}[Y]],$$

where $\mathbb{E}_X$ denotes the expectation taken with respect to the marginal density of X and $\mathbb{E}_{Y|X}$ denotes the expectation taken with respect to the conditional density of Y given X .

Proof. We have that

$$\begin{aligned} \mathbb{E}_X[\mathbb{E}_{Y|X}[Y]] &= \int \left(\int y f_{Y|X}(y|x) dy \right) f_X(x) dx \\ &= \int \int y f_{Y|X}(y|x) f_X(x) dy dx \\ &= \int y \left(\int f_{X,Y}(x, y) dx \right) dy \\ &= \int y f_Y(y) dy = \mathbb{E}[Y] \end{aligned}$$

□

Additionally, for any function $f(\cdot)$, observe that $\mathbb{E}[f(X)Y | X = x] = f(x)\mathbb{E}[Y | X = x]$.

The conditional expectation $\mathbb{E}[Y | X]$ is the *best predictor* of Y given X under squared loss.

Theorem 6 (Conditional expectation as best predictor). *Suppose $\mathbb{E}[Y^2] < \infty$. Then, for any function $g(\cdot)$,*

$$\mathbb{E}[(Y - \mathbb{E}[Y | X])^2] \leq \mathbb{E}[(Y - g(X))^2].$$

Proof. Adding and subtracting $\mathbb{E}[Y | X]$, we have that

$$\begin{aligned}\mathbb{E}[(Y - g(X))^2] &= \mathbb{E}[(Y - \mathbb{E}[Y | X] + \mathbb{E}[Y | X] - g(X))^2] \\ &= \mathbb{E}[(Y - \mathbb{E}[Y | X])^2] + 2\mathbb{E}[(Y - \mathbb{E}[Y | X])(\mathbb{E}[Y | X] - g(X))] \\ &\quad + \mathbb{E}[(\mathbb{E}[Y | X] - g(X))^2].\end{aligned}$$

The cross term is zero: by iterated expectations and the fact that $\mathbb{E}[f(X)Y | X] = f(X)\mathbb{E}[Y | X]$,

$$\mathbb{E}[(Y - \mathbb{E}[Y | X])(\mathbb{E}[Y | X] - g(X))] = \mathbb{E}_X[(\mathbb{E}[Y | X] - g(X)) \mathbb{E}_{Y|X}[Y - \mathbb{E}[Y | X]]] = 0,$$

since $\mathbb{E}_{Y|X}[Y - \mathbb{E}[Y | X]] = \mathbb{E}[Y | X] - \mathbb{E}[Y | X] = 0$. Because $\mathbb{E}[(\mathbb{E}[Y | X] - g(X))^2] \geq 0$, the result follows. $\square$

3.2 Independence

Random variables X and Y are said to be *independent* if $f_{Y|X}(y|x) = f_Y(y)$ for all $x \in \mathbb{R}$, i.e. if the marginal pdf of Y equals the conditional pdf of Y given $X = x$ for all $x \in \mathbb{R}$. Note that $f_{Y|X}(y|x) = f_Y(y)$ if and only if $f_{X,Y}(x,y) = f_X(x)f_Y(y)$.

We note the following two useful properties of independent random variables:

1. If X and Y are independent, then $g(X)$ and $f(Y)$ are also independent for any functions $g : \mathbb{R} \rightarrow \mathbb{R}$ and $f : \mathbb{R} \rightarrow \mathbb{R}$.
2. if X and Y are independent, then $\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y]$.

Indeed, to show the second property, observe that

$$\begin{aligned}\mathbb{E}[XY] &= \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} xy f_{X,Y}(x,y) dx dy \\ &= \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} xy f_X(x) f_Y(y) dx dy \\ &= \int_{-\infty}^{+\infty} x f_X(x) dx \int_{-\infty}^{+\infty} y f_Y(y) dy \\ &= \mathbb{E}[X]\mathbb{E}[Y]\end{aligned}$$

3.3 Covariance

For any two random variables X and Y we can define covariance as

$$\text{Cov}(X, Y) = \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])].$$

As an exercise, check that $\text{Cov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y]$.

Covariances have several useful properties:

1. $\text{Cov}(X, Y) = 0$ whenever X and Y are independent

2. $\text{Cov}(aX, bY) = ab\text{Cov}(X, Y)$ for any random variables X and Y and any constants a and b
3. $\text{Cov}(X + a, Y) = \text{Cov}(X, Y)$ for any random variables X and Y and any constant a
4. $\text{Cov}(X, Y) = \text{Cov}(Y, X)$ for any random variables X and Y
5. $|\text{Cov}(X, Y)| \leq \sqrt{V(X)V(Y)}$ for any random variables X and Y
6. $V(X + Y) = V(X) + V(Y) + 2\text{Cov}(X, Y)$ for any random variables X and Y
7. $V(\sum_{i=1}^n X_i) = \sum_{i=1}^n V(X_i)$ whenever $X_1, \dots, X_n$ are independent

To prove property 5, consider random variable $X - aY$ with $a = \text{Cov}(X, Y)/V(Y)$. On the one hand, its variance $V(X - aY) \geq 0$. On the other hand,

$$\begin{aligned} V(X - aY) &= V(X) - 2a\text{Cov}(X, Y) + a^2V(Y) \\ &= V(X) - 2(\text{Cov}(X, Y))^2/V(Y) + (\text{Cov}(X, Y))^2/V(Y) \end{aligned}$$

Thus, the last expression is nonnegative as well. Multiplying it by $V(Y)$ yields the result.

The correlation of two random variables X and Y is defined by

$$\text{Corr}(X, Y) = \text{Cov}(X, Y) / \sqrt{V(X)V(Y)}.$$

By property 5 as before, $|\text{Corr}(X, Y)| \leq 1$. If $|\text{Corr}(X, Y)| = 1$, then X and Y are linearly dependent, i.e. there exist constants a and b such that $X = a + bY$.

4 Normal Random Variables

Finally, let us introduce the definition of a *multivariate normal distribution*. Let Σ be a positive definite $n \times n$ matrix.² Let μ be $n \times 1$ vector. Then $X \sim N(\mu, \Sigma)$ if X is continuous and its pdf is given by

$$f_X(x) = \frac{\exp(-(x - \mu)^T \Sigma^{-1} (x - \mu)/2)}{(2\pi)^{n/2} \sqrt{\det(\Sigma)}}$$

for any $n \times 1$ vector x .

The multivariate normal distribution has many useful properties that we list here:

1. If $X \sim N(\mu, \Sigma)$, then $\Sigma_{ij} = \text{Cov}(X_i, X_j)$ for any $i, j = 1, \dots, n$ where $X = (X_1, \dots, X_n)'$;
2. if $X \sim N(\mu, \Sigma)$, then $\mu_i = \mathbb{E}[X_i]$ for any $i = 1, \dots, n$;
3. if $X \sim N(\mu, \Sigma)$, then any subset of components of X is normal as well. In particular, $X_i \sim N(\mu_i, \Sigma_{ii})$;
4. if X and Y are jointly normal and uncorrelated, then X and Y are independent;

²Remember that the $n \times n$ matrix Σ is positive definite if $a' \Sigma a > 0$ for any non-zero $n \times 1$ vector a .

5. if $X \sim N(\mu_X, \sigma_X^2)$, $Y \sim N(\mu_Y, \sigma_Y^2)$, and X and Y are independent, then $X + Y \sim N(\mu_X + \mu_Y, \sigma_X^2 + \sigma_Y^2)$;
6. Any linear combination of normals is normal. That is, if $X \sim N(\mu, \Sigma)$ is an $n \times 1$ dimensional normal vector, and A is a fixed $k \times n$ full-rank matrix with $k \leq n$, then $Y = AX$ is a normal $k \times 1$ vector: $Y \sim N(A\mu, A\Sigma A^T)$.

4.1 Conditional distribution

Another useful property of a normal distribution is that its conditional distribution is normal as well. If

$$X = \begin{bmatrix} X_1 \\ X_2 \end{bmatrix} \sim N\left(\begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix}, \begin{bmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{bmatrix}\right)$$

then $X_1|X_2 = x_2 \sim N(\tilde{\mu}, \tilde{\Sigma})$ with $\tilde{\mu} = \mu_1 + \Sigma_{12}\Sigma_{22}^{-1}(x_2 - \mu_2)$ and $\tilde{\Sigma} = \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21}$. If X_1 and X_2 are both random variables (as opposed to random vectors), then $\mathbb{E}[X_1|X_2 = x_2] = \mu_1 + \text{Cov}(X_1, X_2)(x_2 - \mu_2)/V(X_2)$. Let us prove the last statement. Let

$$\Sigma = \begin{bmatrix} \sigma_{11} & \sigma_{12} \\ \sigma_{12} & \sigma_{22} \end{bmatrix}$$

be the covariance matrix of 2×1 normal random vector $X = (X_1, X_2)^T$ with mean $\mu = (\mu_1, \mu_2)^T$. Note that $\Sigma_{12} = \Sigma_{21} = \sigma_{12}$ since $\text{Cov}(X_1, X_2) = \text{Cov}(X_2, X_1)$. From linear algebra, we know that $\det(\Sigma) = \sigma_{11}\sigma_{22} - \sigma_{12}^2$ and

$$\Sigma^{-1} = \frac{1}{\det(\Sigma)} \begin{bmatrix} \sigma_{22} & -\sigma_{12} \\ -\sigma_{12} & \sigma_{11} \end{bmatrix}.$$

Thus the pdf of X is

$$f_X(x_1, x_2) = \frac{\exp\{-(x_1 - \mu_1)^2\sigma_{22} + (x_2 - \mu_2)^2\sigma_{11} - 2(x_1 - \mu_1)(x_2 - \mu_2)\sigma_{12}\}/(2\det(\Sigma))\}}{2\pi\sqrt{\det(\Sigma)}},$$

and the pdf of X_2 is

$$f_{X_2}(x_2) = \frac{\exp\{-(x_2 - \mu_2)^2/(2\sigma_{22})\}}{\sqrt{2\pi\sigma_{22}}}.$$

Note that

$$\frac{\sigma_{11}}{\det(\Sigma)} - \frac{1}{\sigma_{22}} = \frac{\sigma_{11}\sigma_{22} - (\sigma_{11}\sigma_{22} - \sigma_{12}^2)}{\det(\Sigma)\sigma_{22}} = \frac{\sigma_{12}^2}{\det(\Sigma)\sigma_{22}}.$$

Therefore the conditional pdf of X_1 , given $X_2 = x_2$, is

$$\begin{aligned}
f_{X_1|X_2}(x_1|X_2 = x_2) &= \frac{f_X(x_1, x_2)}{f_{X_2}(x_2)} \\
&= \frac{\exp\{ -[(x_1 - \mu_1)^2 \sigma_{22} + (x_2 - \mu_2)^2 \sigma_{12}^2 / \sigma_{22} - 2(x_1 - \mu_1)(x_2 - \mu_2) \sigma_{12}] / (2 \det(\Sigma)) \}}{\sqrt{2\pi} \sqrt{\det(\Sigma) / \sigma_{22}}} \\
&= \frac{\exp\{ -[(x_1 - \mu_1)^2 + (x_2 - \mu_2)^2 \sigma_{12}^2 / \sigma_{22}^2 - 2(x_1 - \mu_1)(x_2 - \mu_2) \sigma_{12} / \sigma_{22}] / (2 \det(\Sigma) / \sigma_{22}) \}}{\sqrt{2\pi} \sqrt{\det(\Sigma) / \sigma_{22}}} \\
&= \frac{\exp\{ -[x_1 - \mu_1 - (x_2 - \mu_2) \sigma_{12} / \sigma_{22}]^2 / (2 \det(\Sigma) / \sigma_{22}) \}}{\sqrt{2\pi} \sqrt{\det(\Sigma) / \sigma_{22}}} \\
&= \frac{\exp\{ -(x_1 - \tilde{\mu})^2 / (2\tilde{\sigma}) \}}{\sqrt{2\pi} \sqrt{\tilde{\sigma}}},
\end{aligned}$$

where $\tilde{\mu} = \mu_1 + (x_2 - \mu_2) \sigma_{12} / \sigma_{22}$ and $\tilde{\sigma} = \det(\Sigma) / \sigma_{22}$. Noting that the last expression equals the pdf of a normal random variable with mean $\tilde{\mu}$ and variance $\tilde{\sigma}$ yields the result.

4.2 Quadratic forms of normal random variables

Recall that a **quadratic form** is a quantity of the form $y' A y$, where A is a symmetric matrix. Suppose that $Z_i \sim N(0, 1)$ i.i.d. for $i = 1, \dots, n$. We already know that $\sum_{i=1}^n Z_i^2 = Z' Z \sim \chi_n^2$. It therefore follows that if $X \sim N_m(\mu, \Sigma)$ and Σ has full rank, then

$$(X - \mu)' \Sigma^{-1} (X - \mu) \sim \chi_m^2.$$

To see this, we now define $Z = \Sigma^{-1/2} (X - \mu) \sim N_m(0, I_m)$, and the result follows from the definition of a χ_m^2 random variable. More generally, it can be shown that for an idempotent matrix M with rank k , $Z' M Z \sim \chi_k^2$.